

Simão Pedro da Silva Vale de Gato

Software Engineer

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MSc in Computer Science and Engineering (Instituto Superior Técnico), specialised in Distributed Systems and Cyber-Physical Systems. Currently working on production Kafka pipelines for enterprise clients, handling around 5M events a day. Published at IEEE VTC 2025 Fall on trajectory prediction for cyclist safety.

Work Experience



Software Engineer @freiheit.com

⌚ 11.2025 - Present 📍 Lisboa, Portugal

Enterprise Logistics Platform Migration 02.2026 - Present

- Designed and built a Java middleware between Kafka and a downstream CRM integration layer, filtering irrelevant events and stashing out-of-order ones for ordered re-emission — cutting load on an overloaded downstream consumer by ~70% and materially improving end-system stability
- Work with high-volume streams: the main topic averages around 5M events a day across 12 partitions
- Ran the production cutover in a 15-minute window, copying consumer offsets and switching consumer group configuration without losing data
- Work on both sides of the integration: Apex classes, triggers and unit tests in Salesforce, and the service that turns Kafka events into HTTP calls to it
- Added integration tests with JUnit and Testcontainers, and built Grafana dashboards and alerts for middleware throughput and failures
- Set up the initial project structure and CI/CD pipelines to the client's engineering standards; run requirements sessions with the client and act as the technical point of contact between them and an external development partner

Dentsply Sirona 11.2025 - 02.2026

- Cut CI pipeline runtime from ~50 to ~30 minutes on Bitbucket Pipelines and GCP by mapping build dependencies, running independent stages in parallel with build groups, and isolating Flutter tests as the main bottleneck (Go + Flutter codebase)
- Built a tool to triage ~100 exposed endpoints and replace a manual security review: it extracts call chains with tree-sitter, runs several independent LLM assessments, then uses a final model to settle disagreements and flag the critical ones



Teaching Assistant @Instituto Superior Técnico

⌚ 01.2025 - 01.2026 📍 Lisboa, Portugal

- Assisted in Operating Systems, Ambient Intelligence (Master's), and Introduction to Architecture of Computers

Education



MSc in Computer Science and Engineering @Instituto Superior Técnico

⌚ 09.2023 - 11.2025 📍 Lisboa, Portugal

- Specialisations: Distributed Systems and Cyber-Physical Systems. **GPA: 17.2/20**
- Thesis published and presented at IEEE VTC 2025 Fall — cyclist safety and trajectory prediction (10.2025). ieeexplore.ieee.org/document/11310081
- Research Grant “BLOCKCHAIN”, funded by the Recovery and Resilience Plan (RRP) and Next Generation EU Funds (07.2025 – 12.2025)



MSc Exchange Semester (Erasmus) @Norges teknisk-naturvitenskapelige universitet

⌚ 08.2024 - 12.2024 📍 Trondheim, Norway

- Parallel Computing, Design of Cyber-Physical Systems, Real-Time Systems, Applied Data Science



• GPA: 15/20

Skills & Qualifications

Programming Languages

Proficient — Java, Python, Go, C

Experienced — Kotlin

Distributed Systems & Data

Apache Kafka (production, high-volume), PostgreSQL / Supabase, schema migrations

Infrastructure, Observability & Testing

Linux, Git / GitHub, Bitbucket Pipelines, Docker (Dockerfiles, Compose), Kubernetes (k9s, Skaffold — running and debugging services), Google Cloud Platform, Grafana (dashboards & alerts), Datadog, JUnit, Testcontainers, Arduino & Raspberry Pi

Enterprise Integration

Salesforce (Apex classes, triggers, unit tests; event-driven integration layers)

Languages

Portuguese (Native), English (B2), French (A1)

Projects

Escala — Volunteer Scheduling Tool @Personal Project

⌚ 2026

- Web app generating balanced volunteer schedules under blocked-date, skill-level, and fairness constraints; serves ~20 volunteers across sound, multimedia, and band teams, replacing a manual scheduling process. Next.js and Supabase (PostgreSQL), with Google authentication and image export for group distribution
- Designed a seven-stage agent delivery pipeline (refinement, adversarial review, implementation, code review, rework, QA, knowledge capture) with specialised subagents selected by file type and bounded retry cycles that escalate to human review; validated output through manual QA, filing follow-up stories for defects found
- github.com/SimaoGato/scheduler

SafeBike — Cyclist Collision Warning System @Master's Thesis

⌚ 2024 - 2025

- Android application in Kotlin using a novel location snapping algorithm and trajectory prediction model; reduced mean spatial error by up to 40% against baseline methods, leading to a paper accepted at [IEEE VTC 2025 Fall](#)

Smart Engagement Room @Academic Project

⌚ 2024

- IoT system using real-time sensors and visual recognition to monitor room engagement

Interests & Hobbies

Technical interests: Home Automations, agent orchestration for software delivery.

Sports: Enjoy running and playing football.

Sound Management: Responsible for managing sound equipment and logistics at my local church.